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THE BESICOVITCH COVERING LEMMA AND MAXIMAL FUNCTIONS

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ABSTRACT. This paper has two purposes. First, we explain and describe the Besicovitch covering lemma, and we provide a new proof. Applications are given, particularly to the ideas of Nagel and Stein about Fatou theorems through approach regions which are not nontangential. Second, we examine the strong maximal function and give a new, simple, geometric proof of its L^p boundedness.

1. Introduction. As usual, let $B(x,r) = \{t \in \mathbb{R}^N : |x-t| < r\}$ be an Euclidean ball. In harmonic analysis, the most commonly used covering lemma is that attributed to Wiener:

Proposition 1.1. *Let K be a compact set in $\mathbb{R}^N$, and let $\mathcal{B} = \{B_\alpha\}_{\alpha \in A}$ be a covering of K by open Euclidean balls. There exist balls $B_\ell \in \mathcal{B}$, $\ell = 1, \ldots, p$, which satisfy these properties*:

(i) *the balls B_ℓ are pairwise disjoint.*
(ii) *Write $B_\ell = B(P_\ell, r_\ell) = \{x \in \mathbb{R}^N : |x - P_\ell| < r_\ell\}$. Then, the three-fold dilates $3B_\ell \equiv \{x \in \mathbb{R}^N : |x - P_\ell| < 3r_\ell\}$ cover K.*

The proof of this proposition is very well known, and we shall not reproduce it here. See [**14,** Chapter 8] or [**21, 25**]. The proposition plays a crucial role in proving weak type $(1,1)$ estimates for the Hardy-Littlewood maximal operator and for other key operators in basic harmonic analysis.

Less well known is the Besicovitch covering lemma (see [**2, 3**]). It has played a role in geometric measure theory (see [**10,** page 145 ff.]) and other geometric contexts. It is attractive since it is a purely geometric

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covering lemma. Measure theory plays no role in its statement or its proof. It can also be used to prove estimates for the Hardy-Littlewood maximal function. But, its flexibility makes it useful in many other contexts. We illustrate some of these in what follows.

2. Statement and proof of Besicovitch. We now present a typical formulation of the Besicovitch result (a much more general version appears in [**10**]).

Proposition 2.1. *Fix an Euclidean space $\mathbb{R}^N$. There exists an integer $M = M(N) > 0$ with the following property: let $\mathcal{B} = \{B_j\}_{j=1}^k$ be a finite collection of open Euclidean balls in $\mathbb{R}^N$ with the property that no ball contains the center of any other. Then, we may write*

$$\mathcal{B} = \mathcal{U}_1 \cup \mathcal{U}_2 \cup \cdots \cup \mathcal{U}_M$$

so that each $\mathcal{U}_j$ is a pairwise disjoint collection of balls, $j = 1, 2, \ldots, M$.

Proof. We begin the proof just as in the traditional proof of the Wiener covering lemma. However, there are some interesting twists at the end.

Set $M = 6^N$. Choose a ball B_1^1 from the collection $\mathcal{B}$ to have the greatest possible radius. If there is more than one such ball, then simply select one.

Next, choose a ball B_2^1 from the remaining balls to have the greatest possible radius and also to be disjoint from B_1^1. If there is more than one, then simply select one.

Continue in this fashion, choosing B_3^1 to be disjoint from B_1^1 and B_2^1 and having the greatest possible radius.

Iteratively select balls until the process stops. It must stop because the original collection $\mathcal{B}$ of balls is finite. Call this collection of balls $\mathcal{U}_1$.

Now, start over again. Choose a ball B_1^2 from the remaining balls which are not in $\mathcal{U}_1$ that has the greatest radius. If there is more than one, then simply select one.

Next, choose a ball B_2^2 from the remaining balls which are not in $\mathcal{U}_1$ that has the greatest radius and is disjoint from B_1^2. If there is more than one, then simply select one.

Continue iteratively choosing balls until the process stops. Call this collection of balls $\mathcal{U}_2$.

Iterate this process, producing pairwise disjoint collections $\mathcal{U}_3$, $\mathcal{U}_4$, etc.

The entire process eventually ends, merely because we began with a finite collection $\mathcal{B}$ of balls. Seeking a contradiction, suppose that we end up with $M+1 = 6^N + 1$ collections $\mathcal{U}_j$.

This means that the first ball B_1^{M+1} in $\mathcal{U}_{M+1}$ intersects a ball from each of the previous $\mathcal{U}_1, \mathcal{U}_2, \ldots, \mathcal{U}_M$. Also, each of those balls from the previous collections has radius greater than B_1^{M+1}. In sum, B_1^{M+1} is touched by M balls, each of radius at least that of B_1^{M+1}. And, none of these balls contains the center of any of the others. But, it is easy to see, either by elementary geometry (see [**15**]), or by calculating volumes, that it is impossible to have M balls with this property. That is a contradiction.

In conclusion, our iterative procedure could only have produced $M = 6^N$ pairwise disjoint families of balls. That completes the proof. Sharp estimates on the size of M appear in [**26**]. □

In what follows, we use the notation $|S|$ to denote the Lebesgue measure of the measurable set S.

Now, for f a locally integrable function on $\mathbb{R}^N$, define

$$Mf(x) = \sup_{R>0} \frac{1}{|B(x,R)|} \int_{B(x,R)} |f(t)|\, dt.$$

This is the *Hardy-Littlewood maximal function.*

In order to implement the Besicovitch covering lemma as we have formulated it above, we need the following technical lemma.

Lemma 2.2. *Let $K \subseteq \mathbb{R}^N$ be a compact set, and let $\mathcal{B} = \{B_\alpha\}_{\alpha \in A}$ be a collection of balls with each $k \in K$ being the center of one of the B_α. Then, there is a subcollection $\mathcal{R} = \{B_j\}_{j=1}^p$ of $\mathcal{B}$ so that no B_j contains the center of any of the other B_k and $\mathcal{R}$ still covers K.*

Proof. See [**10**] for the idea behind this proof.

We claim that the following is true. There is a subcollection $\mathcal{R}$ of balls in $\mathcal{B}$ satisfying the conditions:

(1) whenever $B(p,r)$ and $B(q,s)$ are distinct elements of $\mathcal{R}$, then

$$|p-q| > r+s.$$

(2) Whenever $B(p,r) \in \mathcal{B}$, then there exists a $B(q,s) \in \mathcal{R}$ so that

$$|p-q| \leq r+s \quad \text{and} \quad s > 3r/4.$$

We consider now the class $\mathcal{S}$ of all those subsets $\mathcal{R}$ of $\mathcal{B}$ which satisfy (1) as well as the new condition:

(3) whenever $B(p,r) \in \mathcal{B}$, either $|p-q| > r+s$ for all $B(q,s) \in \mathcal{R}$ or else $|p-q| \leq r+s$ and $s > 3r/4$ for some $B(q,s) \in \mathcal{R}$.

Partially order $\mathcal{S}$ by inclusion. Clearly, $\varnothing \in \mathcal{S}$, and the union of every nest contained in $\mathcal{S}$ is a member of $\mathcal{S}$. Thus, the Hausdorff maximality principle implies that $\mathcal{S}$ has a maximal member $\mathcal{R}$. This will be the collection of balls that we seek.

We note first of all that $\mathcal{R}$ satisfies (2) since, otherwise, (3) would imply that

$$\mathcal{K} \equiv \mathcal{B} \cap \{B(p,r) : |p-q| > r+s \quad \text{for all } B(q,s) \in \mathcal{R}\} \neq \varnothing.$$

As a result, we could choose a ball $B(p',r')$ with

$$\frac{4}{3}r' > \sup\{r : B(p,r) \in \mathcal{K}\}.$$

This would imply that $\mathcal{R} \cup \{B(p',r')\} \in \mathcal{S}$, and that would contradict the maximality of $\mathcal{R}$. □

Proposition 2.3. *The operator M is of weak type $(1,1)$. This means that, for any $\lambda > 0$,*

$$|\{x : Mf(x) > \lambda\}| \leq \frac{C}{\lambda} \cdot \|f\|_{L^1}.$$

Proof. Let

$$E_\lambda = \{x : Mf(x) > \lambda\}.$$

Let K be a compact subset of E_λ. We shall show that

$$|K| \leq \frac{C}{\lambda} \cdot \|f\|_{L^1}.$$

Now, if $k \in K$, then there exists an $R_k > 0$ such that

$$\frac{1}{|B(k, R_k)|} \int\limits_{B(k,R_k)} |f(t)|\, dt > \lambda.$$

However, this just says that

$$|B(k, R_k)| < \frac{1}{\lambda} \cdot \int\limits_{B(k,R_k)} |f(t)|\, dt.$$

Certainly, the collection of balls $\{B(k, R_k)\}_{k \in K}$ covers K, and there is a ball centered at each $k \in K$. Now, invoke the preceding lemma to extract a collection $\mathcal{B}$ of balls which still covers K but so that none of the balls in $\mathcal{B}$ contains the center of any of the others.

However, now, of course, we can invoke the Besicovitch covering lemma to write

$$\mathcal{B} = \mathcal{U}_1 \cup \mathcal{U}_2 \cup \cdots \cup \mathcal{U}_M$$

with each $\mathcal{U}_j$ being a collection of pairwise disjoint balls.

Now, we calculate

$$\begin{aligned} |K| &\leq \sum_{B \in \mathcal{B}} |B| = \sum_{j=1}^{M} \sum_{B \in \mathcal{U}_j} |B| \\ &\leq \sum_{j=1}^{M} \sum_{B \in \mathcal{U}_j} \frac{1}{\lambda} \int_B |f(t)|\, dt \\ &\leq \sum_{j=1}^{M} \frac{1}{\lambda} \|f\|_{L^1} \leq \frac{M}{\lambda} \cdot \|f\|_{L^1}. \end{aligned}$$

We conclude, by the inner regularity of the measure, that

$$|E_\lambda| \leq \frac{M}{\lambda} \cdot \|f\|_{L^1}.$$

That completes the proof. □

3. Generalizations of Besicovitch. The discussion in [**10**] gives a very natural generalization of our Proposition 2.3 to directionally limited metric spaces. We do not need such generality here, so instead restrict attention to Euclidean considerations.

The key fact about balls that is used in the proof of Proposition 2.1 is enunciated in the penultimate paragraph of the proof, namely, we need to know that there is an upper bound on the number of disjoint balls of radius r that can touch a fixed ball of radius r. Thus, it is clear that, if we replace balls by regions U that are comparable to balls, then the result will still hold.

In the next section, the generalized approach regions that we consider (generalizations of classical nontangential approach regions) will give rise to regions U as in the preceding paragraph.

4. Applications to the Fatou theorem. Recall the notation

$$\mathbb{R}^{N+1}_+ \equiv \{(x, y) : x \in \mathbb{R}^N, y \in \mathbb{R}, y > 0\}.$$

The classical Fatou theorem (see [**9**]) states the following. For $\alpha > 1$ and $x \in \mathbb{R}^N$, define

$$\Gamma_\alpha(x) = \{(t, y) \in \mathbb{R}^{N+1}_+ : |x - t| < \alpha y\}.$$

This is, of course, the classical nontangential approach region (below, we also call these α-standard regions). See [**14,** Chapter 8], [**25**].

Theorem 4.1. *Fix $\alpha > 1$. Let f be an L^1 function on $\mathbb{R}^N$. Define $u = P_y * f$, where P_y is the Poisson kernel for the upper half space $\mathbb{R}^{N+1}_+$. Then, for almost every $x \in \mathbb{R}^N$,*

$$\lim_{\Gamma_\alpha(x) \ni (t,y) \to x} u(t, y) = f(x).$$

It has been the traditional wisdom that the nontangential approach is the optimal means of obtaining these boundary limits. See [**16, 19**] for relevant examples.

In the notable paper [**18**], Nagel and Stein show that, in fact, there are tangential approach regions for which a version of Fatou's theorem is still true. In particular, what is essential for a successful approach region $\mathcal{U}$ is *not* that it be nontangential in shape, but rather that the

level sets

$$\mathcal{U}_y = \{x : (x, y) \in \mathcal{U}\}, \quad y > 0,$$

have N-dimensional measure comparable to y^N.

The proof of this result entails following the philosophy of Stein, proving a suitable maximal function estimate. And that, in turn, requires a covering lemma. It is our purpose here to show that the Besicovitch covering lemma is very well suited to this task.

5. Notation and terminology of Nagel-Stein and Mair-Singman. In this section, we recall some basic ideas from the papers **[17, 18]**. We will use most of the notation from **[17]**. We first consider the generalized approach regions for the Fatou theorem.

Definition 5.1. Let $\Omega \subseteq \mathbb{R}^{N+1}_+$ be an open set. Let $\alpha > 0$. For each $t > 0$, we define

$$\Omega(t) \equiv \{x \in \mathbb{R}^N : (x, t) \in \Omega\},$$

and

$$Q(t) = \bigcup_{x \in \Omega(t)} B(x, t).$$

Note that $\Omega(t)$ and $Q(t)$ are both subsets of $\mathbb{R}^N$, which should be thought of as the boundary of the upper halfspace

$$\mathbb{R}^{N+1}_+ \equiv \{(x, y) : x \in \mathbb{R}^N, y \in \mathbb{R}, y > 0\}.$$

Definition 5.2. For $\alpha > 0$ and $y \in \mathbb{R}^N$, we define the *α-standard region* with vertex y to be

$$\Gamma_\alpha(y) = \{(x, t) \in \mathbb{R}^{N+1}_+ : |x - y| < \alpha t\}.$$

In other contexts, these regions have been called the *nontangential approach regions*. See, for instance, **[14,** Chapter 8].

Definition 5.3. Let $\Omega \subseteq \mathbb{R}^{N+1}_+$ and let $\alpha > 0$. We say that Ω is *α-admissible* if

(a) $(0, 0) \in \overline{\Omega}$.

(b) If $0 < s < t$, then $\Omega(s) \subseteq \Omega(t)$.

Now, [**18,** Lemma 1] states the following.

Lemma 5.4. *Let $\alpha > 0$. Assume that Ω is α-admissible. Then, the following properties are obtained*:

(a) *For each $t > 0$,*

$$Q(t) \subseteq \Omega([(\alpha+1)/\alpha] \cdot t).$$

(b) *For each $t > 0$,*

$$|B(0,t)| \leq |Q(t)| \leq C \cdot [(\alpha+1)/\alpha]^N |B(0,t)|.$$

(c) *There exists an integer $P = P(\alpha) > 0$ such that, for any $t > 0$, there exist points $x_1, x_2, \ldots, x_P \in \Omega(t)$ such that $Q(t) \subseteq \cup_{j=1}^{P} B(x_j, 3t)$.*

We shall not reproduce the proof here, but instead refer the reader to [**18**]. We note that, in view of the remarks in Section 3 above, part (c) of the last lemma is of particular interest.

Definition 5.5. Let $\Omega \subseteq \mathbb{R}_+^{N+1}$ have $(0,0)$ as a limit point. For each regular Borel measure μ on $\mathbb{R}^N$ and $y \in \mathbb{R}^N$, we define

$$M_\Omega \mu(y) = \sup_{(x,t)\in\Omega} \frac{\mu(B(y+x,t))}{|B(0,t)|},$$

$$M_\Omega^* \mu(y) = \limsup_{\Omega \ni (x,t) \to (0,0)} \frac{\mu(B(y+x,t))}{|B(0,t)|},$$

$$\mathcal{N}\mu(y) = \sup_{t>0} \frac{\mu(y+Q(t))}{|Q(t)|}.$$

Now, Lemma 5.4 implies the following.

Lemma 5.6. *If Ω is α-admissible and μ is a regular Borel measure on $\mathbb{R}^N$, then*

$$M_\Omega^* \mu(y) \leq M_\Omega \mu(y) \leq C \left(\frac{\alpha+1}{\alpha} \right)^N \mathcal{N}\mu(y)$$

for all $y \in \mathbb{R}^N$.

Now, our key result in pursuit of a generalized Fatou theorem is the following.

Lemma 5.7. *Let Ω be α-admissible. Then, there is a constant $C > 0$ such that, for any finite Borel measure μ on $\mathbb{R}^N$,*

$$|\{x \in \mathbb{R}^N : \mathcal{N}\mu(x) > \lambda\}| \leq \frac{\|\mu\|_1}{\lambda}$$

for all $\lambda > 0$.

Remark 5.8. The lemma shows, in particular, that the operator $\mathcal{N}$ is of weak type $(1,1)$. The operator is trivially of strong type (∞, ∞). Hence, the Marcinkiewicz interpolation theorem implies that $\mathcal{N}$ is of strong type (p,p) for $1 < p < \infty$. As we shall see below, this will show us by standard arguments that the Poisson integral of an L^p function has a boundary limit at almost every point $x \in \mathbb{R}^N$ through approach regions Ω (translated to each x).

Proof of Lemma 5.7. Lemma 5.4 establishes, in a concrete and quantitative fashion, that the regions $\{y+Q(t)\}$ are comparable to Euclidean balls. Hence, as discussed in Section 3, the Besicovitch covering theorem is valid for these new regions. Then, following the proof of Proposition 2.3, we obtain Lemma 5.7. □

Now, it is standard (see [**14,** Chapter 8], [**17, 18**]) to apply Lemma 5.7 to prove the following.

Theorem 5.9. *Let $\alpha > 0$. Let Ω be an α-admissible region. Let $f \in L^p(\mathbb{R}^N)$, $1 \leq p \leq \infty$. Set $u(x) = P_y * f(x)$. Then, for almost every $x \in \mathbb{R}^N$,*

$$\lim_{\Omega \ni (t,y) \to (0,0)} u(x+t, y) = f(x). \tag{5.1}$$

6. The several complex variables situation. Let $B \subseteq \mathbb{C}^n$ be the unit ball. It is a matter of some interest to do harmonic analysis on ∂B. But, we now know that the correct geometry on the boundary is nonisotropic. The proper balls to consider are

$$\beta_2(P, r) = \{z \in \partial B : |1 - z \cdot \overline{P}| < r\}.$$

Here,

$$z \cdot \overline{P} \equiv \sum_{j=1}^{n} z_j \overline{P}_j.$$

In fact, it is convenient to think of these balls as coming from the metric on ∂B, given by

$$\rho(z, \zeta) = \sqrt{|1 - z \cdot \overline{\zeta}|}.$$

The nontrivial argument to show that this is a metric can be found in [**13**].

Due to what we just said about the provenance of the balls (and because they come from a metric, and, since the surface measure of $\beta_2(P, r)$ is about $C \cdot r^n$), it is easy to see that there is an upper bound on the number of disjoint balls of radius r that can touch a fixed ball of radius r. Thus, the Besicovitch theory applies, and we can obtain a covering lemma. Therefore, we can prove the boundedness of the Hardy-Littlewood maximal operator.

One should contrast the approach outlined here with the rather difficult proof of an analogous fact (in a rather more general context), presented in [**22**].

The works [**1, 6, 7**] treat versions of the Nagel-Stein theorem in $\mathbb{C}^n$.

7. Differentiation of integrals and approximations to the identity. The classical Lebesgue theorem on differentiation of integrals states the following.

Theorem 7.1. *Let f be a locally integrable function on $\mathbb{R}^N$. Then, for almost every $x \in \mathbb{R}^N$,*

$$\lim_{r \to 0} \frac{1}{|B(x, r)|} \int\limits_{B(x,r)} f(t)\, dt = f(x),$$

see [**21**].

There is interest in determining whether the balls in this result can shrink to x in a more general fashion. The following is an instance of a result of this kind.

Proposition 7.2. *Let f be a locally integrable function on $\mathbb{R}^N$. Then, for almost every $x \in \mathbb{R}^N$,*

$$\lim_{r\to 0} \frac{1}{|B(x+r,r)|} \int_{B(x+r,r)} f(t)\,dt = f(x). \tag{7.1}$$

Proof. We prove this result by noting that the ball $B(x+r,r)$ is comparable to the ball $B(x,2r)$. Thus, the Besicovitch covering result applies, and the corresponding maximal function is bounded as usual. Then, the result (7.1) follows by standard techniques. □

In a related vein, we may consider approximations to the identity. Let $\varphi \in C_c^\infty(\mathbb{R}^N)$ have these properties:

(a) $\operatorname{supp}\varphi \subseteq B(0,2)$;
(b) $0 \le \varphi \le 1$;
(c) $\int \varphi(x)\,dx = 1$.

Define $\varphi_\epsilon(x) = \epsilon^{-N}\varphi(x/\epsilon)$. The following is a standard fact (for which, see **[21]**).

Theorem 7.3. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$. Then:*

$$\lim_{\epsilon\to 0} f * \varphi_\epsilon(x) = f(x) \tag{7.2}$$

for almost every $x \in \mathbb{R}^N$.

There is interest in knowing in what more general ways the functions φ_ϵ can be allowed to converge weak-$*$ to the Dirac delta mass and still obtain a positive result in the nature of (7.2). Next is a sample of such a result.

Proposition 7.4. *Let $f \in L^1_{\mathrm{loc}}(\mathbb{R}^N)$. Then:*

$$\lim_{\epsilon\to 0} \int_{\mathbb{R}^N} f(x+\epsilon-t)\varphi_\epsilon(t)\,dt = f(x)$$

for almost every x.

The proof of this result is handled similarly to the preceding proposition.

Certainly, results of the type discussed here are considered in [**17, 18**].

8. Combinatorics. The Besicovitch theorem has some surprising applications in diverse parts of mathematics. For example, in the paper [**8**], the result is applied to studying the most efficient way to pack coins in the plane (coins of different denominations, and of different radii). The techniques described here do not give a precise description of the optimal arrangement of the coins (but, see [**26**] for sharp estimates). However, they give estimates on the number of coins in any given region.

It is clear that a host of similar problems could be studied using this technique. For instance one could study sphere packing, both for spheres all of the same size and for spheres of varying sizes, in higher dimensions. The Kepler sphere-packing problems in dimension 3 was, of course, solved by Hales in [**11**].

9. The strong maximal function. The strong maximal function is defined as follows.

Definition 9.1. Let $f \in L^1_{\text{loc}}(\mathbb{R}^N)$. Define

$$M^* f(x) = \sup_R \frac{1}{|R|} \int_R |f(t)|\, dt,$$

where R ranges over rectangular boxes (with arbitrary side lengths) with sides parallel to the axes. It is important to note here that the boxes R are *not* necessarily cubes. They can have arbitrary eccentricity.

It is known (see [**23**]) that M^* is *not* of weak type $(1,1)$ (just calculate M^* of the Dirac delta mass), but it *is* bounded on L^p for $1 < p \leq \infty$. The standard method for proving this last assertion is to majorize M^* by the iteration of N one-dimensional maximal functions of Hardy-Littlewood type (see the original paper [**12**]). The proof is so brief and informative that we include it here.

Theorem 9.2. *The strong maximal function is bounded on L^p for $1 < p \leq \infty$.*

Proof. The case $p = \infty$ is trivial; thus, we concentrate on $1 < p < \infty$.

Now, for $f \in L^p(\mathbb{R}^N)$,

$$\begin{aligned} M^*f(x) &= \sup_R \frac{1}{|R|}\int_R |f(t)|\,dt \\ &= \sup_R \Bigg(\frac{1}{|R_1|}\cdot\frac{1}{|R_2|}\cdot\cdots\cdot\frac{1}{|R_N|} \\ &\qquad \times \int_{R_1}\int_{R_2}\cdots\int_{R_N} |f(t_1,t_2,\ldots,t_N)|\,dt_N\cdots dt_2\,dt_1\Bigg). \end{aligned}$$

Here, $R = R_1 \times R_2 \times \cdots \times R_N$. This last is

$$\begin{aligned} &\le \sup_{R_1}\left(\int_{R_1}\sup_{R_2}\left(\int_{R_2}\cdots\sup_{R_N}\left(\int_{R_N}|f(t_1,t_2,\ldots,t_N)|dt_N\right)\cdots dt_2\right)dt_1\right) \\ &\le M_1M_2\cdots M_Nf(x). \end{aligned}$$

Of course, M_j denotes the classical Hardy-Littlewood maximal function in the single jth variable. We know that each of these functions is bounded on L^p, so it follows that M^* is bounded on $L^p(\mathbb{R}^N)$. □

This last method of proof was not considered to be completely satisfactory since it obviates the geometry of the situation. Fortunately, Cordoba and Fefferman [**5**] found another, more geometric way to prove the L^p boundedness of the strong maximal function. The strong maximal function does not lend itself well to analysis with the Besicovitch lemma, although some partial results are possible.

We shall now offer a more elementary and more direct geometric proof. For the proof, we shall exploit the Stein-Weiss interpolation theorem [**24**]. For simplicity, we restrict attention to $\mathbb{R}^2$, although it is easy to see that the ideas work as well in $\mathbb{R}^N$ for any N.

Let $E \subseteq \mathbb{R}^2$ be any bounded, measurable set. Let $D = D(0,R)$ be the smallest closed disc, centered at the origin, that contains E. We may as well assume that $R > 1$. Then,

$$M^*\chi_E(x) \le M^*\chi_D(x).$$

Therefore, let us think about $M^*\chi_D$.

It is easy to see that, for $|x| \ge 5R$, we have

$$M^*\chi_D(x) \le C \cdot \max((|x|-2R)^{-1}, R^2\cdot(|x|-2R)^{-2}).$$

It follows that, for $p > 2$, $M^*\chi_D(x) \in L^p$, and, in fact,

$$\begin{aligned}
\|M^*\chi_D\|_{L^p} &\leq C \int_{|x|\leq 5R} 1^p \, dx^{1/p} + C \int_{5R}^{\infty} (t-2R)^{-p} \, t \, dt^{1/p} \\
&\qquad + C \int_{5R}^{\infty} (t-2R)^{-2p} R^{2p} \, t \, dt^{1/p} \\
&\leq C R^{2/p} + C \int_{5R}^{\infty} (t-2R)^{-p+1} \, dt^{1/p} \\
&\qquad + C \, R^2 \int_{5R}^{\infty} (t-2R)^{-2p+1} \, dt^{1/p} \\
&= C \, R^{2/p} + C \left(\frac{(t-2R)^{-p+2}}{-p+2} \Big|_{5R}^{\infty} \right)^{1/p} \\
&\qquad + C \, R^2 \left(\frac{(t-2R)^{-2p+2}}{-2p+2} \Big|_{5R}^{\infty} \right)^{1/p} \\
&= C \, R^{2/p} + C \frac{(3R)^{2/p-1}}{(p-2)^{1/p}} + C \, R^2 \frac{(3R)^{2/p-2}}{(2p-2)^{1/p}} \\
&\leq C_p \, (R^{2/p} + R^{2/p-1} + R^{2/p}) \leq C_p \, R^{2/p} \\
&= C_p \, \|\chi_D\|_{L^p}.
\end{aligned}$$

The conclusion, then, is that $\|M^*\chi_E\|_{L^p}$ is finite. This is true for each characteristic function of each bounded, measurable set E. By the closed graph theorem [**4**], we may conclude that M^* maps characteristic functions boundedly to L^p, $p > 2$. It is noteworthy here that the closed graph theorem applies to more general operators than linear ones.

Thus, M^* is of strong type (p, p) when it acts on characteristic functions. Hence, it is restricted of weak type (p, p) (this means that the operator is of weak type (p, p) when restricted to functions that are characteristic functions of measurable sets, see [**24**]). It is also obviously of strong type (∞, ∞). By the Stein-Weiss interpolation theorem, we may conclude that M^* is of strong type (p, p) for $2 < p \leq \infty$.

We can now linearize the maximal function in a standard fashion as follows: let

$$x \mapsto R_x$$

be a measurable assignment of rectangles (with sides parallel to the axes) to points x of $\mathbb{R}^N$. Then, the operator

$$f \mapsto \frac{1}{|R_x|} \int_{R_x} f(t)\, dt \tag{9.1}$$

is linear. And, the strong maximal function is majorized by the supremum of all of these linear operators (9.1), with the supremum taken over all possible measurable assignments of rectangles.

Next, we may apply a duality argument to each of these linearized operators and pass to the supremum to conclude that M^* is of strong type (p, p) for $1 < p < 2$. Thus, we know that M^* is bounded on all L^p strictly between 1 and ∞ except for $p = 2$. Finally, we may apply interpolation one more time to conclude that the maximal function M^* is bounded on L^2, that is, the main result about the strong maximal function.

10. Concluding remarks. The remarkable theorem of [**20**] tells us that the converse of the proof of Theorem 5.9 is true, namely, if, for a particular Ω and $1 \leq p \leq 2$, we know that the limit (5.1) exists almost everywhere for $f \in L^p(\mathbb{R}^N)$, then the corresponding maximal operator $\mathcal{N}$ is of weak type (p, p).

Nagel and Stein showed that their version of the Fatou theorem is sharp. Fausto di Biase [**1, 6, 7**] produced various sharp forms of the Nagel-Stein theorem in the several-complex-variable context.

The philosophy espoused by these theorems suggests an entirely new point of view in harmonic analysis, one that deserves further exploration in future papers.

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